



A phenomenological approach on the influence of paramagnetic iron in cement stone on 2D T_1 - T_2 relaxation in single-sided ^1H nuclear magnetic resonance

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ABSTRACT

In recent years, there has been an increasing interest in ^1H nuclear magnetic resonance (NMR) relaxometry to obtain information on the hydration and porosity of cement stone. By spin-spin (T_2), spin-lattice (T_1), and combined relaxation studies, new insights were gained on liquid and crystal bound water as well as pore water exchange. A conversion of relaxation times to pore sizes has been developed and shown to give consistent results. In this study, we use single-sided ^1H NMR to obtain two-dimensional T_1 - T_2 relaxation distributions on systematically varied cement stone samples as well as hydrated and solid cement phases. Results show a richness in features beyond past studies and are interpreted qualitatively in terms of water found in chemically different pore environments. Both T_1 and T_2 surface relaxivity of calcium silicate surrounded liquid water linearly correlate with the cement's Fe_2O_3 content. Results are discussed based on a theory of interacting surface relaxation mechanisms.

1. Introduction

1.1. Nuclear magnetic resonance relaxation

Liquids, such as water, can strongly interact with a surrounding pore surface. Effects are not only macroscopically observable as capillary suction or condensation but also through ^1H nuclear magnetic resonance (NMR) relaxation [1,2]. Here, overall NMR relaxation is dominated by the relaxation of molecules in direct contact to the pore walls [3,4]. In the Brownstein-Tarr fast-diffusion model [3], the relaxation rate $1/T_{1,2}$ is represented as the sum of bulk and surface relaxation terms:

$$\frac{1}{T_{1,2}} = \frac{1}{T_{1,2 \text{ bulk}}} + \frac{S}{V} \cdot \frac{\epsilon}{T_{1,2 \text{ surface}}} \quad (1)$$

where $1/T_{1,2}$ is the relaxation rate of (1) spin-lattice or (2) spin-spin (in 1/s) interaction, $1/T_{1,2 \text{ bulk}}$ is the relaxation rate of the bulk liquid (in 1/s), S is the pore wall surface (in m^2), V is the pore volume (in m^3), ϵ is the thickness of a surface layer, usually assumed to be the size of a single water molecule (0.28 nm), and $T_{1,2 \text{ surface}}$ is the relaxation rate of hydrogen in this surface layer in direct contact with the pore wall. Based on the assumptions (i) $T_{1,2 \text{ bulk}} \gg T_{1,2 \text{ surface}}$ and (ii) a simple, rescalable, and self-similar pore geometry, the measured relaxation rate

can be directly converted to a pore size by multiplying with the surface relaxivity λ . Bryar et al. [5] suggested “that surface relaxivity is constant for each pure mineral and that observed surface relaxivity parameters are weighted averages of these inherent relaxivities”.

Especially for cementitious materials, the pore wall relaxation processes are greatly influenced by paramagnetic impurities, such as iron and manganese [5,6]. According to Korringa, Seever, and Torrey [7], the relaxation rate can be represented by the sum of bulk relaxation and the individual contributions of paramagnetic and non-magnetic species, weighed by their proportion of occupied surface sites and the average surface site residence times [6]. Usually, these different relaxation influences are omitted since a single contribution, mostly by paramagnetic impurities, is said to dominate relaxation. As shown in simple powder experiments [5,8] as well as theoretically described [9,10], the surface relaxation rate $1/T_{1,2 \text{ surface}}$ is proportional to the concentration c of paramagnetic impurities, such as Fe_2O_3 :

$$\frac{1}{T_{1,2 \text{ surface}}} \propto c(\text{Fe}_2\text{O}_3) \quad (2)$$

Relaxation processes become considerably more complex in the case of experiments in inhomogeneous magnetic fields, which usually is the case for single-sided NMR. Since spins in different magnetic field strengths precess with different frequencies, an almost instantaneous

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dephasing of spins, and thus, a loss of signal, is seen after the very first exciting pulse. The free induction decay (FID) is usually not observable [2]. Luckily, dephasing caused by a constant difference in magnetic field strength can be countered by the well-known Carr-Purcell Meiboom-Gill (CPMG) pulse sequence [11,12] and thus, allows the measurement of T_2 , spin-spin or transversal relaxation. Time-dependent fluctuations of the magnetic field as arise from random movement of spins inside the otherwise time-invariant constant magnetic field gradient as well the temporary proximity to paramagnetic impurities [8], still cause irreversible NMR relaxation. Besides T_2 relaxation, the T_1 , spin-lattice or longitudinal relaxation can be registered. For measuring T_1 relaxation, a saturation recovery pulse sequence is best used in single-sided NMR [2]. Here, first several perturbing 90° -pulses are applied to destroy the magnetization, followed by a recovery period and a subsequent CPMG pulse sequence. The CPMG signal is usually integrated to increase signal quality, though it otherwise allows a combined T_1 and T_2 relaxation measurement. The less than ideal interaction between the excitation coil and inhomogeneous magnetic field in single-sided NMR experiments tend to complicate the picture further. With any given pulse, not all addressed spins respond accordingly [2,13,14]. Over the three-dimensional sensitive volume, both the static magnetic field as well as the coil excitation strongly varies and thus, a variety of flip angles appears after any pulse. The caused deviations in excitation and signal response need to be accounted for as further addressed below. In the following, we refer to T_1 and T_2 relaxation times and the T_1/T_2 relaxation time ratio simply as “ T_1 ”, “ T_2 ”, and “ T_1/T_2 ”, respectively. Furthermore, Fe_2O_3 content is abbreviated by simply “ Fe_2O_3 ”.

1.2. ^1H NMR relaxation on cementitious materials

Systematic and extensive research on the hydration and desorption of cement pastes enabled the attribution of T_2 to different pore water species [15–17]. Liquid water ranging between 0.1 ms for calcium silicate interlayer water and 10 ms for water in larger capillary pores was found to become best visible using a CPMG sequence. The extremely fast relaxation of solid components of only around 0.02 ms was measured by a two-pulse solid-echo sequence, detecting both solid and liquid water. Differentiation of the two was achieved through the different relaxation behaviour in the FID [17] with solid bound water following a Gaussian relaxation behaviour [18,19] and liquid water following exponential behaviour. By drying cement stone, a shortest liquid T_2 relaxation time of 0.075 ms was found [15,20] and interpreted as the relaxation of a single layer of water on the inner pore surface, thus giving $T_{2,\text{surface}}$. Dividing an assumed layer thickness ε of 0.28 nm by this relaxation time, a direct conversion from relaxation time to pore size through a surface relaxivity λ of 3.73 nm/ms has been found for white Portland cement (WPC) stone at 20 MHz containing 0.3 to 0.4 wt % of Fe_2O_3 and a w/c ratio of 0.4 [15,20]. With this conversion, values for pore sizes have been received that matched the existing estimations [20,21] and models [22] well. Such simple and direct conversion from relaxation time to pore size provides an easy way of probing cementitious pore size distributions using ^1H NMR and thus, is highly desirable.

Most of the recent relaxation studies on cementitious samples concerned T_2 measurements with the CPMG sequence. This measurement is significantly faster and due to the close correlation with T_1 in liquids gives almost the same richness in information. Even though the combination of T_1 and T_2 measurements provide even deeper insight, only few studies exist combining T_1 and T_2 relaxation. Bortolotti et al. [23] used a closed magnet setup with an inversion-recovery (for T_1) pulse sequence measuring the subsequent FID (for T_2). They reported pore species using T_1 comparable to those found using T_2 before. Deviations appeared for solid contents, having a much longer T_1 than T_2 , as reported before [4]. Using the combined information, they were able to identify two solid components with a short T_1 (SS) and a long T_1 (SL), the latter again being comprised of two separate signals, as plotted in

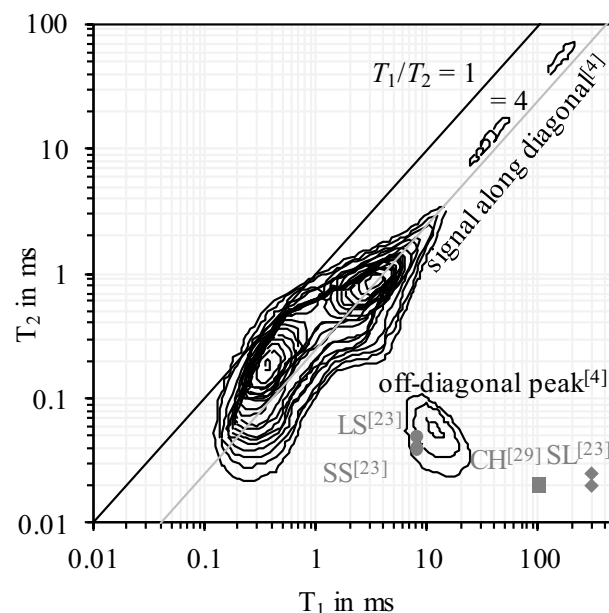


Fig. 1. Results from T_1 - T_2 measurements obtained from (i) McDonald et al. [4] (a WPC stone, age 4 days at 20 MHz) showed as equally spaced up, contour-like black lines with two main features as diagonal signal and an off-diagonal peak, (ii) from Schreiner et al. [29] (OPC stone, cured for 42 days at 19 MHz) showed as grey square, interpreted as calcium hydroxide (CH), and (iii) Bortolotti et al. [23] (WPC stone with added NaCl, age 10 days at 20 MHz), depicted as grey dots for liquid short (LS), solid short (SS) and solid long (SL) signals. T_1 were partly averaged from relaxation time distributions and corresponding T_2 were partly chosen based on an educated guess, due to the limited information reported. The black and grey lines indicate $T_1/T_2 = 1$ and 4, respectively.

Fig. 1. They assigned their signal to portlandite and ettringite for SL as well as less rigidly bound hydroxide for SS, the latter being accompanied by a liquid component (LS) with similar T_1 , thus being assigned to hydroxide, associated with calcium ions. An even more interesting way of combining T_1 and T_2 relaxation are so-called two-dimensional (2D) studies [4,24–26]. Such 2D studies are not as widely used because most information is obtainable by faster T_2 relaxation measurement alone and furthermore, the evaluation involves complex 2D numerical inversion. Though not wholly mathematically correct [27], such inversion is usually generously referred to as an inverse Laplace transformation (ILT). Song et al. [25] introduced the use of a T_1 - T_2 2D-ILT using an NMR-frequency of 2 MHz finding all major signal features on a diagonal with a T_1/T_2 of between 1 and 4 for different sand- and limestones. Later, McDonald et al. [4] were able to show a similar context on white cement stones and an NMR-frequency of 20 MHz including a strong, off-diagonal peak, as shown in Fig. 1. In combination with T_2 - T_2 relaxation measurements, this was interpreted as an exchange of water between pores of different size, rather than signal from solid bound water. Similar results were obtained on ordinary Portland cement (OPC) stone [24] incorporating higher amounts of paramagnetic impurities, though the off-diagonal peak was only visible after the first day and furthermore, T_1/T_2 was shifted to a value of 2. A very recent study [28] on 2D T_1 - T_2 measurements found comparable and even further peaks as before, though simply assigning these to noise in the measurement. 2D measurements were even expanded to the use of single-sided NMR devices [24], though the features observed were not at all as rich in detail.

1.3. The occurrence of iron in cement

Paramagnetic impurities, such as iron, hugely influence relaxation effects and thus, knowing where and how these iron phases are found

and bound is important. Upon hydration, calcium aluminate clinker (C_3A)¹ first reacts with sulphates forming ettringite needles (tri-sulphate, Aft) visible on a micrometre scale. Such needles are formed by aluminate rich columns with sulphates bond in between [30]. Later, these converge to (ettringite) mono-sulphate (AFm), depending on the sulphate supply in the pore solution [31]. Furthermore, hydrogarnets (CAH or CASH) may be formed. It is originally believed that the iron-bearing Brownmillerite (C_4AF) reacts comparably, hydrating to iron ettringite as well as iron hydrogarnet ($C(A,F)H$ or $C(A,F)SH$) [31]. First doubts have been raised by Stark et al. [32] reporting that, though pure Brownmillerite alone reacts accordingly, these products were never found in hydrated Portland cements. Instead, only aluminate leached iron-rich clinker residues were found. Recently, a comprehensive study on the “fate of iron” during hydration [33,34] showed that Brownmillerite first forms ferrihydrite (amorphous or poorly crystalline $Fe(OH)_3$ [34]) and only later and considerably slower forms iron silica hydrogarnet. Calculations suggested that only these phases and no ferrite tri- or mono-sulphates are formed in Portland cement.

1.4. Boundaries of NMR relaxation distributions and limitations of measurement

Based on the information given in the literature, some boundaries and connections for T_1 , T_2 , and their ratio can be derived somewhat simplifying the interpretation of 2D T_1 - T_2 distribution results. The spanned relaxation distribution space is illustrated in Fig. 2. First, the shortest possible T_2 is limited by the rigid lattice relaxation [35], giving a lower limit for T_2 of 0.02 ms [17,23,35]. T_1 is not limited in the long regime and both liquids as well as solids show long T_1 . In between, T_1 shortens considerably for viscous substances and the lower limit for T_1 as suggested by the Bloembergen-Purcell-Pound (BPP) equations [35] is around 0.5 ms and might even become smaller when affected by paramagnetic impurities [4,36].

Concerning limits in T_1/T_2 ratio, no T_2 longer than its corresponding T_1 exists. Thus, the diagonal $T_1 = T_2$ limits its ratio to the space below 1. For rigidly bound water the BPP equations suggest an increase of T_1 while T_2 further shortens, severely increasing T_1/T_2 . For the case of liquids interacting with pore surfaces, the fast-diffusion model [3] has to be considered. Based on the surface chemistry [5,37] and the amounts of paramagnetic impurities on the wall surfaces [4], T_1 surface and T_2 surface are defined and thus, a T_1/T_2 ratio. The latter theoretically being independent of paramagnetic impurities [4] but strongly increasing with increasing surface adsorption strength [37] and wettability [38,39]. A decrease in surface to volume ratio comes with an increase in spherical pore diameter and with it a movement upward along the parallel line to the diagonal $T_1 = T_2$. Thus, T_1/T_2 stays constant and signal only deviates from this line when the relaxation process is not governed by the surface but rather bulk [25] or diffusive relaxation [40,41] and T_1/T_2 is de- or increased, respectively. Based on the fast-diffusion model, two differently sized pores can be discriminated, when they are well separated. With increased connectivity, water molecules tend to exchange in the timeframe of the experiment and the formerly well-separated signals merge until they appear as one average sized pore [42]. In the intermediate region of connectivity, pore exchange becomes visible [42]. In T_1 - T_2 distributions, these appear as two additional corners to the peaks along the diagonal, forming a square. Since exchange affects both T_1 and T_2 relaxation at the same rate, for an increased T_1/T_2 ratio, peaks in T_1 relaxation merge before T_2 . By this, the square is deformed and ultimately collapses to a vertical line in T_1 - T_2 distributions [43]. The one corner peak has been mathematically predicted [42,44] and experimentally shown [45] to be negative and as an exception, might even be found above the $T_1 = T_2$ diagonal. Due to the non-negativity constraint, past ILT algorithms

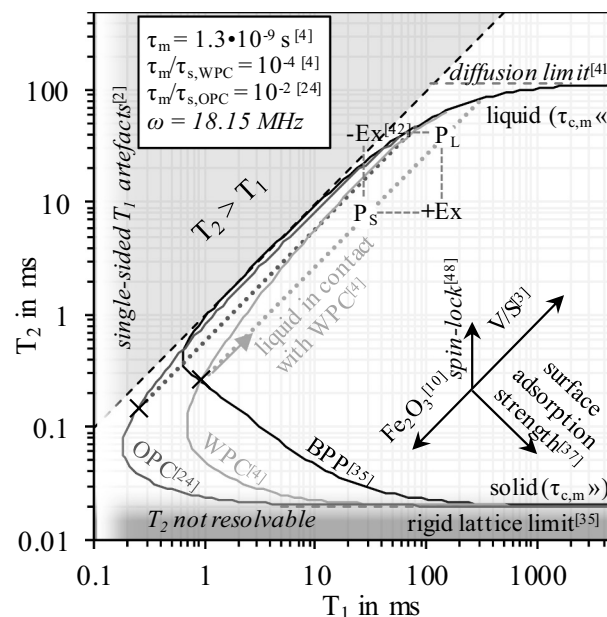


Fig. 2. T_1 and T_2 distribution relations and limits. NMR device and measurement specific limits are written in *italic*, material or NMR specific in normal letters. The black line is based on the Bloembergen-Purcell-Pound (BPP) equations [35] and indicates the expected T_1 , T_2 , and T_1/T_2 for hydrogen different rigidly bound from liquid ($\tau_c \ll$) to solid ($\tau_c \gg$). Dark and light shaded solid lines show surface relaxations based on the work of Korb, McDonald, and Co-workers [4,9,10] for WPC [4] and OPC [24] stone with parameters as shown in the inset box. From these surface relaxation lines, dotted lines indicate the expected region of signal for liquid water in differently sized pores as predicted from the fast diffusion model [3]. P_L and P_S indicate larger and smaller pores with -Ex and +Ex being corresponding negative and positive exchange peaks, arranged in a square [42]. T_2 is limited between the diffusion limit [41] and the rigid lattice limit [35]. The compass rose type arrows indicate the direction of relaxation time changes due to the named influences.

could not account for these and only in recent years, new algorithms have been developed for this matter [46,47].

Besides the boundaries for the existence of T_1 - T_2 relaxation pairs, limitation of the actual resolution of relaxation times for the devices used can be addressed. First, for single-sided NMR, the longest T_2 is limited by the diffusive relaxation in the magnetic field gradient. Based on the parameters used in this study, T_2 is limited by $T_{2,\text{Diff}} = 115$ ms for bulk water. Second, it appears reasonable that very short T_2 (approximately below the echo spacing T_E in CPMG) become difficult to resolve [24]. Furthermore, pulses in single-sided NMR can only incompletely excite and flip spins [14]. Even after careful calibration, some spins remain aligned to the magnetic field after the perturbing pulses [2]. This causes a vertical offset in 1D T_1 relaxation experiments and in 2D T_1 - T_2 relaxation experiments evaluated by 2D-ILT, generates signal artefacts at the shortest T_1 accounted for in the evaluation. Last, for T_2 relaxation measurements in heterogeneous magnetic fields, other than in constant fields, the echo spacing needs to be kept constant and short [48]. Otherwise, relaxation processes are further complicated by diffusive contributions or fast relaxing signals cannot be registered. Unfortunately, the overall induced pulse intensity alters and increases the registered T_2 , bringing it closer to its corresponding T_1 in the rotating frame due to so-called spin-locking affects [14,49,50]. This only mildly effects relaxation where $T_1 \sim T_2$, though is becoming more severe with increasing T_1/T_2 . For a very similar single-sided NMR device as used in this study, an increase of the T_2 measured of 20% was observed [50].

In this study, we present results from single-sided 1H nuclear magnetic resonance 2D T_1 - T_2 relaxation experiments of different solid and hydrated phases, found in cement stone, as well as cement stones with

¹ In the following, cement oxide notation is written in *italic*.

Table 1
Cement composition.

Measurement	Parameter	CEM I 52.5 R (WPC)	CEM I 52.5 R (OPC)
		wt% of dry cement mass	
Carbon sulphur analyser (sample dried at 105 °C)	CO ₂	3.00	1.95
	SO ₃	3.50	3.40
X-ray fluorescence (XRF) (after ignition)	Na ₂ O	0.11	0.26
	MgO	0.76	0.74
	Al ₂ O ₃	4.30	3.70
	SiO ₂	22.6	22.0
	SO ₃	3.77	3.36
	K ₂ O	0.81	0.47
	CaO	67.0	65.8
	TiO ₂	0.13	0.21
	MnO	0.03	0.04
	Fe ₂ O ₃	0.17	1.30
	Loss on ignition	4.5	3.1
	Alite	60.3	69.4
	Belite	17.6	8.1
X-ray diffraction (XRD) (dried at 40 °C)	Aluminate-phases	7.0	10.1
	Brownmillerite	0.4	1.2
	Quartz	0.0	0.6
	Arcanite	1.3	0.0
	Anhydrite	4.3	0.0
	Hemihydrate	1.9	2.6
	Gypsum	0.1	3.6
	Calcite	4.9	3.4
	Portlandite	2.2	0.9

varied iron-content. The results overcome limitations of past single-sided 2D studies and even show a depth of detail, comparable to results from closed magnet setups. We evaluate our results based on a theory of interacting relaxation mechanisms.

2. Experiments

2.1. Materials

In the experiments, two different cements according to DIN EN 197-1:2014 were used: An ordinary Portland cement (OPC) and a white Portland cement (WPC). Table 1 gives the cements' main constituents. According to the manufacturer, Blaine-values were 5800 cm²/g and 5400 cm²/g for WPC and OPC, respectively. Furthermore, to obtain cement with a gradually changing amount of iron, WPC and OPC were combined in mass ratios of 75:25, 50:50, and 25:75, mixed thoroughly, and subsequently used as if received in this way. Samples are named after their calculated Fe₂O₃ content as g_{Fe2O3}/g_{dry cement} in wt%. The OPC and WPC were chosen to be well comparable in its oxide composition, primarily showing differences in iron content (expressed as Fe₂O₃). The approach of cement preparation is further addressed in the discussion. Phase composition as measured by X-ray diffraction (XRD) shows differences in the type of calcium silicate phases with WPC having higher amounts of belite as well as the type of sulphate setting regulator with WPC containing anhydrite instead of gypsum in OPC. Stoichiometric calculations suggest that Brownmillerite in OPC is underestimated by XRD and an amount of ~3.8 wt% is more realistic.

Hardened cement stone was made by thoroughly mixing cement powder with tap water in a w/c-mass ratio of 0.5 and filling the fresh paste into a 250 ml polyethylene screw-cap bottle. The filled bottle was vibrated and then sealed using a rubber plug avoiding any air being trapped inside. Bottles were continuously rotated for 24 h in an overhead-shaker to prevent sedimentation [51,52]. After 24 h, the rubber plug was removed, the remaining space on top was filled with saturated calcium hydroxide solution, and bottles were closed with a screw-cap. Hardened cement stone was stored inside the bottle at laboratory climate (~22 °C) for a duration of 9 months. Discs with 60 mm diameter

and a thickness of 10 mm were wet cut from the bottles' bottoms and were saturated in calcium hydroxide solution to prevent leaching. Samples showed an average mass increase of 0.5 wt%. Furthermore, a sample of WPC stone was dried at 40 °C inside a ventilated oven until reaching a constant mass ($\Delta m \leq 0.01$ wt%/d) [53] and subsequently stored inside a glass desiccator above silica beads. In advance of any NMR measurement, samples were wrapped in thin plastic foil to prevent evaporation or uptake of moisture.

Second, hydrated cement phases were investigated. High purity synthetic Alite (C₃S) [54], low-iron white aluminate cement (WAC) (Ternal white [55], Kerneos SA), as well as synthetic ettringite powder (CASUL, Remondis GmbH) were used. The dry powders were mixed with deionized water in a w/c-ratio of roughly 0.5 and cured inside a small plastic petri dish each. The samples were kept moist and measured at an age of 7 and 28 days. Furthermore, CASUL powder was compacted inside a small plastic petri dish, mixed with deionized water, and measured instantaneously, since no further hydration reaction was expected.

Last, powders of solid phases typically found in hydrated cement were investigated. Commercial calcium hydroxide, CASUL, commercial hydrotalcite, and commercial gypsum plaster were measured. The plaster was mixed in a water/solid ratio of 0.46, thoroughly cured, and subsequently ground. The powders were dried at 105 °C for calcium hydroxide and 40 °C for ettringite and gypsum, then compacted inside a small plastic petri dish and levelled flush. Its bulk densities were determined by its weight, divided by the beforehand determined volume of the petri dishes. The main constituents of synthetic Alite, WAC, CASUL, and gypsum are reported in the supplement.

2.2. Characterisation of cement stone

To obtain NMR independent information on the pore structure and chemical composition of the OPC and WPC stones, measurements were conducted by mercury intrusion porosity (MIP), thermogravimetric analysis (TGA), and XRD. For MIP measurements, samples were dried at 40 °C in a ventilated chamber and measured on an AutoPoreIV, Micromeritics, in incremental mode. Samples were crushed to pieces of roughly 5 mm. For data evaluation, surface tension and contact angle of mercury were set to 0.485 N/m and 140° [56], respectively. For XRD and TGA characterisation, the sample was prepared by solvent exchange precisely following the instructions described elsewhere [57]. TGA measurements were conducted on a TGA 4000, Perkin Elmer, with 50 mg sample mass under nitrogen atmosphere from 30 to 900 °C at a heating rate of 10 K/min. Contents of Portlandite and calcite were evaluated tangentially as described elsewhere [56]. For XRD measurements, Rutile, Kronos 2220, was added as internal standard at a level of 20 wt% of the sample and the two were mixed thoroughly. Measurements were executed using an X-ray powder diffractometer X'Pert Pro, Panalytical, with an X'Celerator detector in a range of 5 to 70° 2 θ with a step size of 0.0167° 2 θ . The samples were analyzed qualitatively, followed by quantification using the Rietveld method.

2.3. Single-sided ¹H nuclear magnetic resonance

NMR measurements were carried out with a single-sided NMR MOUSE PM5 (Nuclear Magnetic Resonance Mobile universal surface explorer Profile Mouse 5 mm), developed and manufactured by the Institute of Macromolecular Chemistry of the RWTH Aachen University [58] using a minispec MQ20 spectrometer, Bruker. In a recent publication, further information has been given on the device [53]. 2D T₁-T₂ experiments were conducted with a saturation recovery sequence beginning with 5 perturbing 90° pulses with decreasing time spacing, followed by a recovery period varied in 51 durations logarithmically spaced between 0.01 and 1000 ms, followed by a CPMG sequence consisting of a total of between 400 and 1000 echoes with an echo time spacing T_E of 0.03 ms. The measurement depth was fixed at 2 mm inside

Table 2
NMR measurement parameters used for cement stone samples.

Sequence	Parameter	Value	Unit
Spectrometer and magnet/coil array	Magnetic field gradient	21.6	T/m
	Measurement frequency	18.15	MHz
Pulse sequence	Number of scans	1024 to 2048	–
Saturation recovery (for T_1)	Number of T_1 recovery periods	51	–
	Durations of T_1 recovery periods	0.01 to 1000	ms
CPMG (for T_2)	Echo-spacing T_E	30	μ s
	Pulse-length	4.5	μ s
	Acquisition window	5	μ s
	Number of T_2 echoes	400 to 1000	–

any sample. Measurement durations were between 3 and 14 h, depending on the number of scans and T_2 echoes. The latter influences the needed recycle delay between scans to prevent heating of coil and sample at short recovery times. All parameters used for cement stone samples, hydrated Alite and WAC, and water-mixed CASUL samples are given in Table 2. For powder samples, the numbers of T_2 echoes and T_1 recovery durations were adjusted as needed.

For data pre-processing, echo amplitudes were blank corrected and normalised by the amplitude of deionized bulk water A_w , measured in a 1D T_2 relaxation experiment with sufficient recovery duration between subsequent scans of 20 s. Furthermore, the first two echoes were corrected for transient effects [2,13] as suggested by the same measurement of water by dividing the first echo by 0.84 and the second by 1.01. Last, a slight “spurious oscillation” [59] was corrected and smoothed by the following empirical function.

$$A_c(i) = A(i) \cdot \left(1 - (-1)^i \cdot A_z \cdot \exp\left(-\frac{t}{T_z}\right) \right) \quad (3)$$

where $A_c(i)$ is the corrected amplitude of echo i , $A(i)$ is the amplitude as measured, i is number of the echo, A_z is the correction amplitude factor, T_z (in ms) is a correction time constant. Based on a large number of samples, A_z and T_z were set to 0.04 and 0.1 ms and held constant for all cement stone sample evaluations. This way, the normalised and corrected signal in the following called Φ_{NMR} is obtained. Φ_{NMR} by definition gives a value of 100%, if the entire sensitive measurement volume is occupied by water and has proven to be a meaningful indicator of a cementitious sample's water filled porosity [53].

Exponential decays were evaluated by numerical inversion, usually referred to as inverse Laplace transformation (ILT). For this, a non-negative constrained algorithm, developed at the University of Wellington, New Zealand [60,61], was used. The basics of ILT as well as the need for some kind of regularization have been widely discussed [2,4,24,25,60–63] and thus, we limit our description to the essentials. For 2D-ILT of cement stone samples, relaxation values were evaluated between 0.01 and 1000 ms for T_1 and between 0.02 and 20 ms for T_2 . A total of $41 \times 61 = 2501$ logarithmically spaced T_1 - T_2 relaxation time pairs were evaluated using a regularization parameter $\alpha = 10^6$, leading to a calculation time of roughly 5 min. For synthetic hydrated Alite and WAC stone samples, relaxation times and resolutions were adjusted as needed.

3. Results

3.1. Composition of hydrated cement stone samples

The use of MIP for gaining pore size distributions has been widely discussed in recent years [64] and it is well known that the choice of drying [21,65] as well as contact angle [21] strongly affect results. We chose the sample preparation and measurement in this comparatively fast and easy way since we merely aim to provide evidence on the comparability of the cement stone's porosity and pore size distributions. Fig. 3 shows MIP intrusion curves of WPC and OPC stones. Results

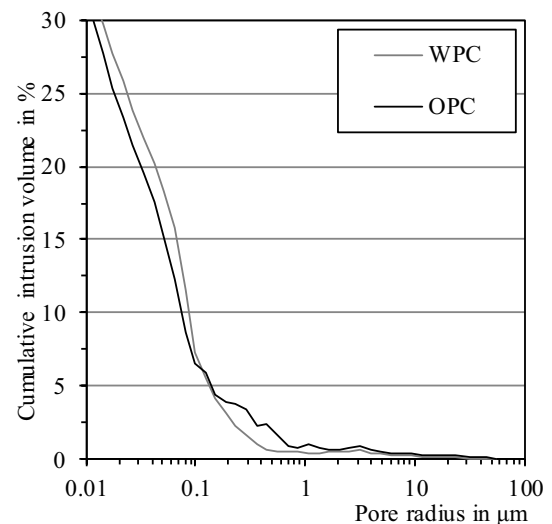


Fig. 3. Results from MIP intrusion of the capillary porosity for WPC and OPC.

indicate some amount of scattering, especially for the OPC sample between 0.1 and 1 μ m. Intrusion volume above 5% intruded volume is not affected and average pore radii in this range only differ by about 20%. The overall pore structures of the OPC and WPC stones used in our study appear to not differ as much, as has been reported in past studies [21,66]. In Table 3, results from TGA and XRD measurements are shown. The cement stone samples were well hydrated, with only traces of clinker and most importantly, no unhydrated Brownmillerite left. Results of other hydrated products show an increase of Portlandite as well as a decrease in ettringite and monocarboaluminate with increasing original Fe_2O_3 in the cement. The latter might well be exchanged with hydrotalcite, since the two phases share most of its spectral features in XRD. The reduced amorphous content of sample 25:75 indicates that resultant phases in this sample are slightly overestimated, probably caused by a small miss match of sample and internal standard. Results from TGA as well as amounts of portlandite and calcite, derived from TGA [56], as shown in Table 3 and visualized in the supplement, mostly match with XRD. TGA indicates even greater amounts of portlandite, though it might be noted that hydrotalcite theoretically has a comparable evaporation temperature [56] and might also contribute to this effect. Loss of free water between 40 and 400 $^{\circ}\text{C}$ decreases with increasing Fe_2O_3 content with the loss between 50 and 200 $^{\circ}\text{C}$ being most prominent. This might well be associated with the reduced amounts of ettringite and monocarboaluminate or hydrotalcite, seen in XRD.

3.2. 2D T_1 - T_2 relaxation distributions of powders of solid cement phases

Fig. 4 shows results from the measurements of water, ettringite, calcium hydroxide, and gypsum. Furthermore, the BPP equations [35] is plotted with added terms for the limits of T_2 due to the rigid lattice $T_{2\text{rl}}$ and diffusional relaxation $T_{2\text{Diff}}$. In the following, results are given in order of amplitude in per cent, T_1 relaxation time in ms, and T_2 relaxation time in ms in the form $A/T_1/T_2$. Water, having $A/T_1/T_2$ of 84.6%/2300 ms/115 ms well complies with the beforehand described theories of NMR relaxation. The vertically elongated peak indicates the diffusive relaxation mechanism of water in the magnetic gradient field and furthermore the influence of spin-locking effects causing a shift of the apparent T_2 , closer to its corresponding T_1 . Unfortunately, an undesired repercussion of the 2D-ILT is illustrated here as well. The signal amplitude of water, though normalised to itself for 1D-ILT, does not give an amplitude of 100%. In Table 4, theoretical water contents (incorporating hydroxides) of all substances are shown together with the acquired signal intensities by single-sided NMR, evaluated by 2D-ILT as

Table 3

Results from XRD and TGA measurement of the cement paste samples in g/100 g hydrated and solvent exchanged binder.

	Mineral phase	100:0 (WPC)	75:25	50:50	25:75	0:100 (OPC)
X-ray diffraction (XRD) (solvent exchange) [57]	Alite	0.83	0.24	0.24	0.26	0.24
	Belite	0.49	0.43	0.17	0.2	0.01
	Aluminate-phases	0.00	0.00	0.00	0.00	0.00
	Brownmillerite	0.00	0.00	0.00	0.00	0.00
	Quartz	0.00	0.01	0.07	0.09	0.14
	Arcanite	0.00	0.01	0.00	0.07	0.01
	Anhydrite	0.00	0.00	0.01	0.02	0.00
	Hemihydrate	0.00	0.00	0.00	0.00	0.00
	Gypsum	0.00	0.00	0.00	0.00	0.00
	Calcite	1.85	1.30	1.16	0.96	0.70
	Portlandite	10.33	10.53	12.00	14.34	13.32
	Ettringite	5.16	4.69	4.29	5.22	4.41
	Monosulfoaluminate	0.00	0.01	0.00	0.03	0.00
	Hemicarboaluminate	0.00	0.00	0.05	0.17	0.08
	Monocarboaluminate	2.91	2.92	2.55	3.1	2.17
	Amorphous	78.43	79.85	79.47	75.53	78.92
Thermo-gravimetric analysis TGA (solvent exchange) [57]	Free Water 40–400 °C (stepwise)	22.90	20.85	20.73	19.71	19.28
	400–500 °C (tangential)	3.37	3.56	3.74	3.95	4.15
	650–750 °C (tangential)	0.66	0.63	0.48	0.31	0.18
	Remnants at 900 °C	68.64	69.99	70.34	71.22	71.66
TGA derived [56]	Portlandite	13.85	14.64	15.38	16.24	17.06
	Calcite	1.51	1.42	1.08	0.73	0.40

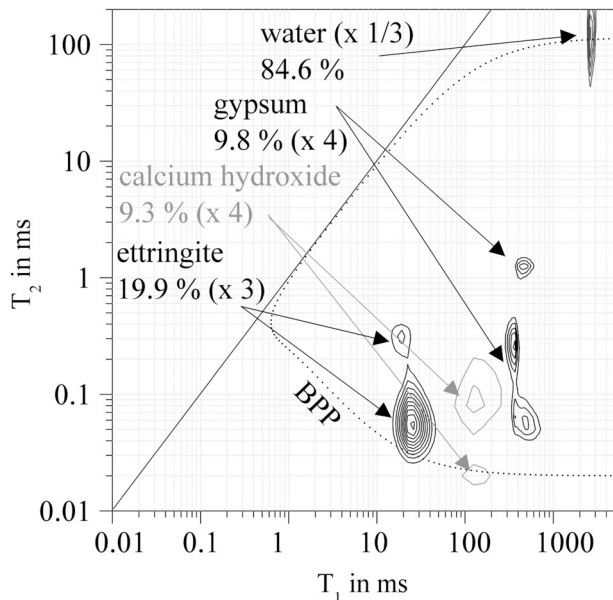


Fig. 4. Combined Results from 2D-ILT of liquid water, ettringite, calcium hydroxide, and gypsum. Results are shown as equally spaced up contour lines after rescaling with the factor in the brackets to account for the hugely different signal intensities. The number below each caption is the original signal's amplitude as integral of the signal peaks in 2D-ILT. The solid black line gives the diagonal $T_1 = T_2$, the dotted line is based on the Bloembergen-Purcell-Pound (BPP) equations for increasing correlation time τ_c with added terms for the limits of T_2 due to the rigid lattice T_{2rl} and diffusional relaxation T_{2-Diff} .

well as by multi-exponential fitting. For the latter, the shortest T_2 relaxation time was set to meet the shortest T_2 , obtained from 2D-ILT. Depending on the regularization parameter α , a reduction of the total amplitude in 2D-ILT of 10 to 20% is observed, thus limiting the further interpretation of amplitudes quantitatively. The signal obtained from ettringite powder gives 20.9%/20 ms/0.075 ms. Based on the 32 water molecules in ettringite and the measured bulk density of the powder, a total signal of about 30% would have been expected. Of all the total of 32 water molecules bound in ettringite, 2 are bound as zeolite water, 24 are water bound on the ettringite column's surface and in between columns, and 6 are bound as hydroxide inside the crystal structure [30].

Table 4

Density, theoretical water content of powder substances based on its bulk density and water molar mass, signal intensity obtained by single-sided ^1H NMR evaluated by multi-exponential fitting and 2D-ILT, and primary T_1 relaxation time.

Substance	Bulk density	Theoretical water content	Signal intensity		T_1
			By multi-exponential fit	By 2D-ILT	
–	g/cm ³	g/cm ³ · 100	%		ms
Water	1.00	100	100	84.6	2300
Ettringite	0.70	30	23.1	20.9	20
Calcium hydroxide	0.66	17	13.9	9.3	150
Hydrotalcite	0.62	22	21.2	15.8	200
Gypsum	1.12	23	12.8	9.8	400

The results suggest that both column bound ettringite as well as zeolite water is seen here. The more closely bound hydroxide is not detected. Furthermore, the small peak at slightly longer T_2 might even indicate a detectable difference in NMR signal between zeolite and column bound water. The peak of ettringite as found here closely resembles the off-diagonal peak in 2D T_1 - T_2 cross-relaxation experiments on cement stone found by McDonald et al. [4] as well as the liquid short (LS) signals found by Bortolotti et al. [23] (compare Fig. 1). Calcium hydroxide shows two peaks with a total signal intensity of 9.3% both at a T_1 relaxation time of 150 ms, though having slightly different T_2 relaxation times of 0.02 and 0.1 ms. Based on the theoretical hydrogen content, a total signal of about 17% would have been expected. The relaxation time of 0.02 ms is considerably beyond the limitations of the used CPMG sequence (compare Fig. 2). Thus, neither the signal intensity nor its T_2 relaxation time can be interpreted quantitatively. Nonetheless, we repeatedly found comparable results including this peak at extremely short T_2 for measurements of calcium hydroxide. In the literature [23,67], a differentiation between hydroxyl and more mobile hydrogen in calcium hydroxide has been made and might best explain the two peaks. When mixed with deionized water (not shown), the latter peak becomes even more pronounced. Results from hydrotalcite shown in the supplementary materials have considerable similarity to calcium hydroxide though with slightly greater total signal intensity. Last, results from gypsum powder give three peaks at different T_2 at a T_1 of

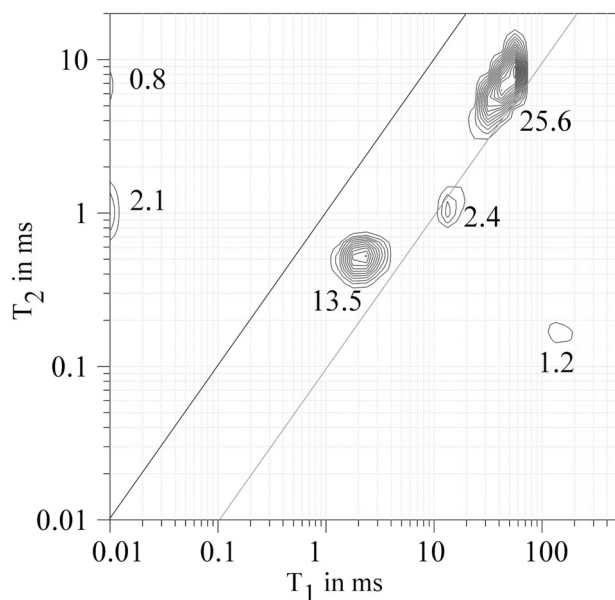


Fig. 5. Synthetic Alite after 7 days of hydration. The black and grey lines indicate the diagonal at $T_1/T_2 = 1$ and 10, respectively. The averaged T_1/T_2 ratio of the three main diagonal peaks is 7.

approximately 400 ms. For gypsum, a T_2 surface relaxation time of 0.2 ms at 23 MHz has been reported [68], well fitting to the centre peak. The peak at $T_2 = 1$ ms is assigned to remnants of gypsum pore water after drying. The peak at 0.05 ms indicates remnants of solid signal. Overall, the greatest discrepancies are found between theoretical water content in the sample and acquired signal for gypsum, indicating that any crystal bound water is mostly missed.

3.3. 2D T_1 - T_2 relaxation distributions of hydrated cement phases

Results of synthetic Alite after 7 days of hydration are shown in Fig. 5 with three major peaks all along a diagonal line with an averaged T_1/T_2 ratio of 7 and a shortest peak at $T_1/T_2 = 2.5$ ms/0.5 ms. According to the existing theories, this signal is emitted from liquid water inside the porous calcium silicate structure with varying pore sizes, strongly influenced by the pore surface relaxivity of calcium silicate. Furthermore, an off-diagonal peak appears at $T_1 \sim 150$ ms, roughly matching the peak of calcium hydroxide at longer T_2 . Last, peaks appear above the $T_1 = T_2$ diagonal at very small T_1 , as discussed further for the cement stone samples. After 28 days of hydration, as shown in the supplement, the ratio of the two peaks change in favour of the faster relaxing one indicating the progressed hydration, though no change on relaxation time is observed. In Fig. 6, results of WAC hydrated for 7 days give a considerably more complex picture. An abundance of well discernible peaks is visible, which appear to be reproducible when comparing the measurement after 7 and 28 days (see supplement). While some seem to follow a parallel to the diagonal $T_1 = T_2$, a number of further peaks emerge. The peaks of these calcium aluminate rich phases all appear around a T_1 of between 1 and 50 ms with a T_1/T_2 of between 10 and 200. Last, for results of water-mixed ettringite, shown in the supplement, the main peak for solid ettringite at 10 ms/0.075 ms remains, while liquid water appears as an elongated ridge to longer relaxation times with partly constant T_1/T_2 of ~ 25 , indicating the well connected interparticle space.

3.4. 2D T_1 - T_2 relaxation distributions of cement stone

Results of mature cement stones with a w/c ratio 0.5 at the age of roughly 9 months are shown in Fig. 7. In the following descriptions, we

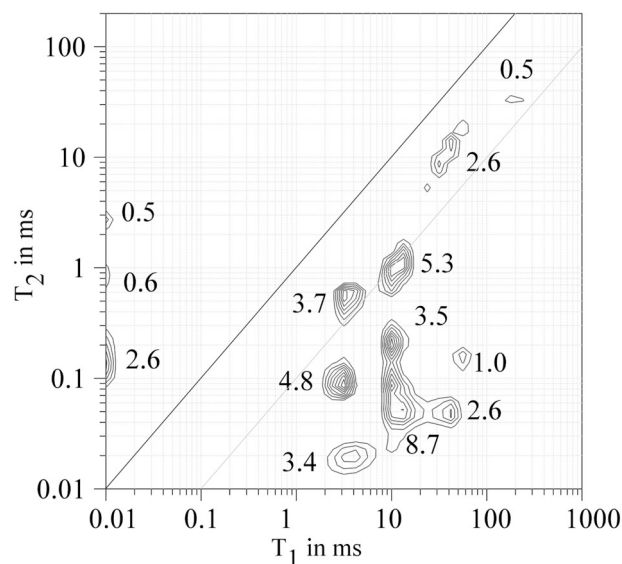


Fig. 6. WAC after 7 days of hydration. The black and grey lines indicate the diagonal at $T_1/T_2 = 1$ and 10, respectively.

focus first on the WPC stone in Fig. 7b). A total number of 8 peaks are found beside the main signal along the diagonal $T_1/T_2 \sim 3$. As reported before for T_1 - T_2 experiments [4], most signal arises along this diagonal line, though absolute numbers of T_1/T_2 ratio decrease with lower measurement frequency [4,35]. According to existing theories, this signal is emitted from liquid water inside the iron-bearing porous calcium silicate structure with varying pore sizes. Furthermore, two peaks appear at $T_1 = 0.01$ ms, mirroring the main peaks along the diagonal. This positive signal with a shorter T_1 than T_2 can physically not exist. We ascribe this effect to the incomplete flipping of spins inside the heterogeneous magnetic field of the single-sided NMR device [2]. Then for the interpretation, these peaks might theoretically simply be added to the signal on the diagonal. At lower T_2 relaxation times, two peaks are registered at 7.1%/10 ms/0.075 ms and 1.4%/200 ms/0.2 ms, closely resembling the peaks of powdered ettringite and calcium hydroxide, respectively. Furthermore, a peak at 1.6%/1000 ms/0.05 ms emerges. Last, small peaks appear at 0.4%/20 ms/0.4 ms, 0.2%/200 ms/8 ms, and 0.2%/1000 ms/20 ms, sharing a common T_1/T_2 ratio of ~ 25 . These small signals, as low as 1/100 of the total signal intensity, were originally simply assigned to noise in the measurement. However, following further measurements and also reports in the literature [28], these very small signals were repeatedly found at re-occurring spots.

In Fig. 7c) to e), results from mixtures of WPC and OPC and in f), results for the OPC stone sample are shown. In general, similar features to those of WPC can be found with some minor differences. The signal along to the diagonal $T_1 = T_2$ lies closer to it, only shifted to $T_1/T_2 \sim 1.5$ [24]. For increasing amounts of Fe_2O_3 , the signal along the diagonal shows some curvature at shortest relaxation, slightly tilting away from the diagonal. More importantly, relaxation times are reduced with increasing Fe_2O_3 . Peaks at $T_1 = 0.01$ ms slightly increase with Fe_2O_3 . This is most likely an effect of the overall reduced T_1 relaxation time, increasing single-sided T_1 artefacts of these samples. With increasing Fe_2O_3 , the peak at 10 ms/0.07 ms is considerably reduced while at the same time, appears to be divided, with a second peak at 1 ms/0.08 ms emerging. Furthermore, a number of smaller peaks appear at $T_1 > 100$ ms at different T_2 .

Last, in Fig. 7a), results of a sample of WPC stone dried at 40 °C are reported. Upon drying, the signal with longer T_2 is lost due to the evaporation of liquid water [20]. The peak at $T_1/T_2 = 10$ ms/0.075 ms on the other hand is only slightly reduced. Furthermore, a peak at very

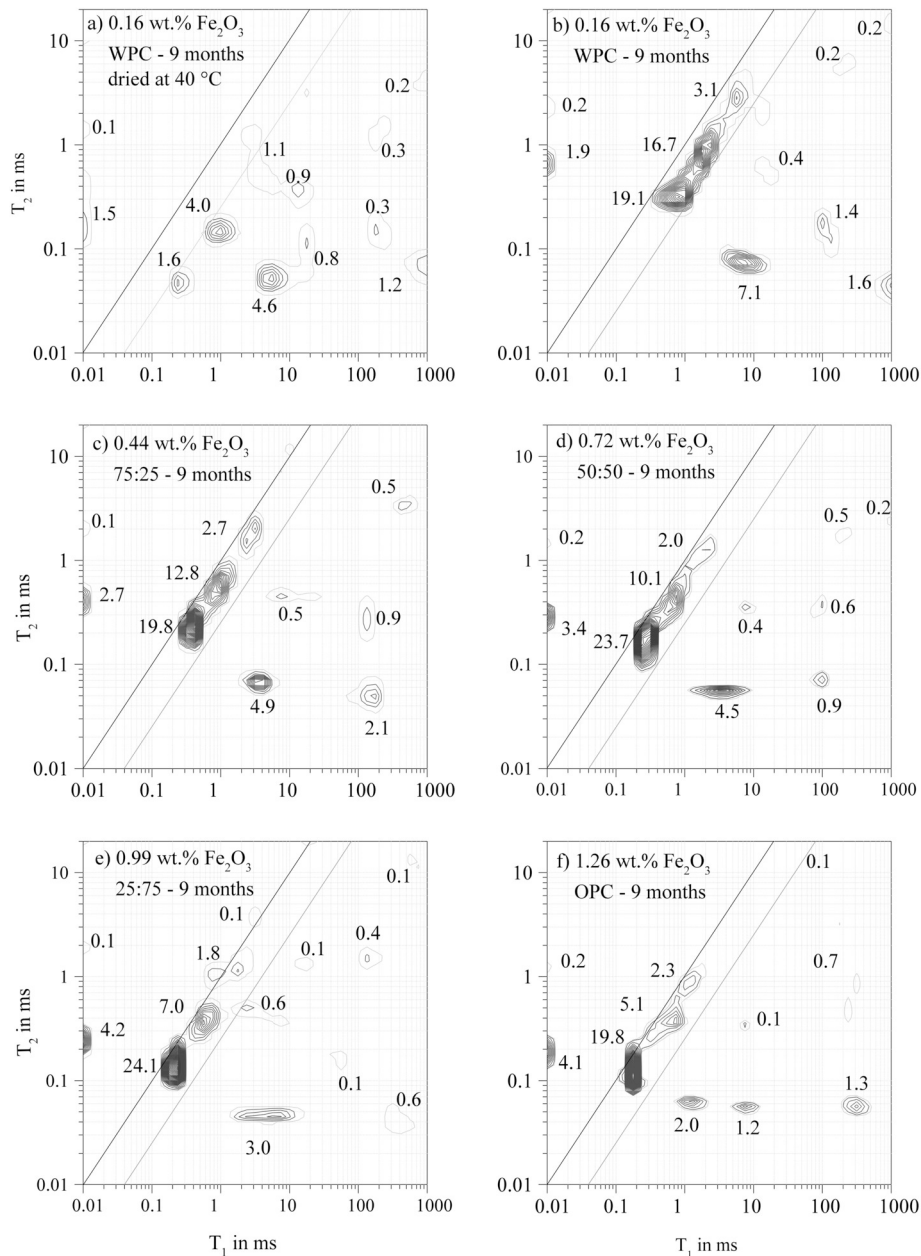


Fig. 7. 2D T_1 - T_2 distributions of cement stone samples a) of WPC dried at 40 °C, b) WPC, c), d), e) mixtures of WPC and OPC with increasing Fe_2O_3 , and f) OPC. Results are shown as equally spaced up black lines with addition of a grey line, indicating any signal above zero. The number beside each peak is its signals amplitude as integral of each peak. The black and grey diagonal lines indicate $T_1/T_2 = 1$ and 4, respectively. Besides signal on a diagonal at $T_1/T_2 = \sim 2$, which presumably is water inside the porous CSH structure, further peaks emerge, discussed in the text.

short relaxation of 0.25 ms/0.075 ms appears. We repeated this experiment and found similar results on separately prepared samples made from the same cement.

3.5. Projection of 2D-ILT and integration of peaks

To illustrate the Fe_2O_3 dependent relaxation behaviour of cement stone further, the complex 2D-ILT relaxation distributions were simplified by partial projection as explained in the following and illustrated in Fig. 8 for saturated WPC stone: First, results were evaluated more smoothly by a decrease of the regularization parameter α to 10^4 . Second, since absolute values do vary with the ILT regularization, as shown above, results were normalised to one. Third and last, signal is separated into three parts: (i) signal from presumably liquid water parallel along the $T_1 = T_2$ line, (ii) signal clearly deviating from that

line with the border at $T_1/T_2 = 10$, and (iii) signal at shortest T_1 accounted for in the evaluation at $T_{1-\min} = 0.01$ ms. The resulting signal was projected in their T_1 and T_2 directions and plotted as cumulative curves beginning at longest $T_{1,2}$.

Fig. 9a) to f) show the projected cumulative curves for signal along the diagonal and off-diagonal signals. For signal along the diagonal, assigned to liquid water, a clear trend of decreasing T_1 and T_2 relaxation times with increasing Fe_2O_3 is observed. Paramagnetic impurities increase surface relaxivity and thus, relaxation. Both for T_1 and T_2 relaxation an almost entirely parallel behaviour of cumulative relaxation curves is found, though T_1 appears to be more influenced by scattering and deviations. The latter may simply be explained by measurement resolution, as for T_1 only 41 relaxation times were evaluated in a range of 5 decades based on only 51 saturation recovery durations, while for T_2 , 61 relaxation times were evaluated in a range of 3 decades based on

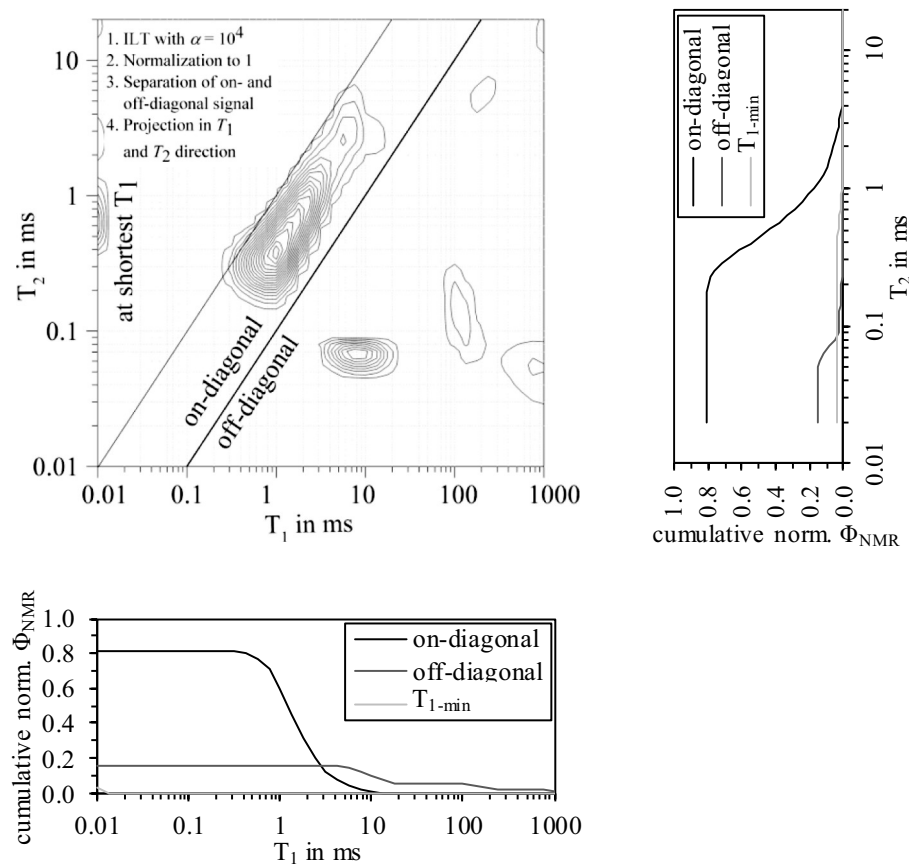


Fig. 8. Visualization of partial projection of signal, along the diagonal line $T_1 \sim T_2$ and for signal, off-diagonal, and signal at shortest T_1 accounted for in the ILT evaluation at $T_{1-\min} = 0.01$ ms.

400 CPMG echoes. To further illustrate the paramagnetic dependent behaviour, relaxation times were read at 0.4 of the total normalised signal, indicated by the dotted lines and black arrows in Fig. 9 and plotted as relaxation rates ($1/T_{1,2}$) in Fig. 10. This particular signal range represents interhydrate and gel water. In the following, we assume that the corresponding pore sizes do not differ between WPC and OPC, as indicated by the MIP results in Fig. 3. In Fig. 10, an almost perfect linear trend is observed for both T_1 and T_2 relaxation rates $1/T_{1,2}$. Since absolute values of relaxation rate depend on surface relaxivity and pore size, and the latter is not known here, absolute values do not provide sensible information. Since, on the other hand, pore sizes are equal for T_1 and T_2 at one point of normalised signal intensity, the ratio of T_1/T_2 is evaluated in the inset of Fig. 10. Based on the linear fits of relaxation rate,² a T_1/T_2 of ~ 5 is found for a Fe_2O_3 free material. For great amounts of Fe_2O_3 , the ratio converges to ~ 1.5 . For off-diagonal signal deviating from $T_1 \sim T_2$, shown in Fig. 9c) and d) some Fe_2O_3 dependence of relaxation times appears though with the OPC sample not following the overall trend. The intensity of these off-diagonal peaks are comparatively low and thus, not unambiguously interpreted. For the signal at shortest T_1 accounted for in the ILT evaluation at $T_{1-\min} = 0.01$ ms shown in Fig. 9e) and f), similar behaviour in T_2 as for the liquid signal can be found, though with increasing total signal amplitude indicating its close relationship to the liquid water signal.

² It might be noted that if the signal components at shortest T_1 , presumably due to incomplete flipping of spins, are uniformly redistributed along the diagonal signal of the same T_2 , the ratios of theoretically Fe_2O_3 free calcium silicate surfaces and increased Fe_2O_3 contents slightly increase and tend towards ~ 7 and ~ 2 , respectively.

This increase in signal is an indication of the overall faster T_1 relaxation in OPC, increasing the artefact at shortest T_1 due to incomplete flipping of spins.

4. Discussion

Theories and models describing NMR relaxation phenomena in porous media are well established and have provided extensive information on pore size and nano-structure. In cementitious samples, however, the overall picture is further complicated by naturally high amounts of paramagnetic impurities, differently bound water fractions as well as further signal components.

4.1. Influence of Fe_2O_3 on NMR relaxation of liquid water in cement stone

The results from 2D T_1 - T_2 measurements reveal predominantly liquid water signal of roughly 80% of the total signal, for all cement stones (see Fig. 9a) and b)). This liquid water signal appears to overall follow the expected behaviour along the diagonal with higher T_1/T_2 for WPC [4] and lower T_1/T_2 for OPC [24] (visible in Fig. 7). Furthermore, results show the Fe_2O_3 dependent reduction of relaxation time (see Fig. 9a) and b) and Fig. 10). The T_1/T_2 ratio for liquid water in cement stone samples is not necessarily uniform, as seen in Fig. 7. With increasing Fe_2O_3 , samples show a greater curvature at shortest T_2 relaxation times. Interestingly, a similar behaviour of curved 2D-ILT distributions had already been visible in past studies of 2D relaxation distributions of cement stones. McDonald and Co-Workers [4,24] reported results from two WPC and one OPC pastes and found strikingly varying behaviour: One WPC showing such curvature (as shown in Fig. 1) along $T_1/T_2 = \sim 4$, a second WPC without such curvature,

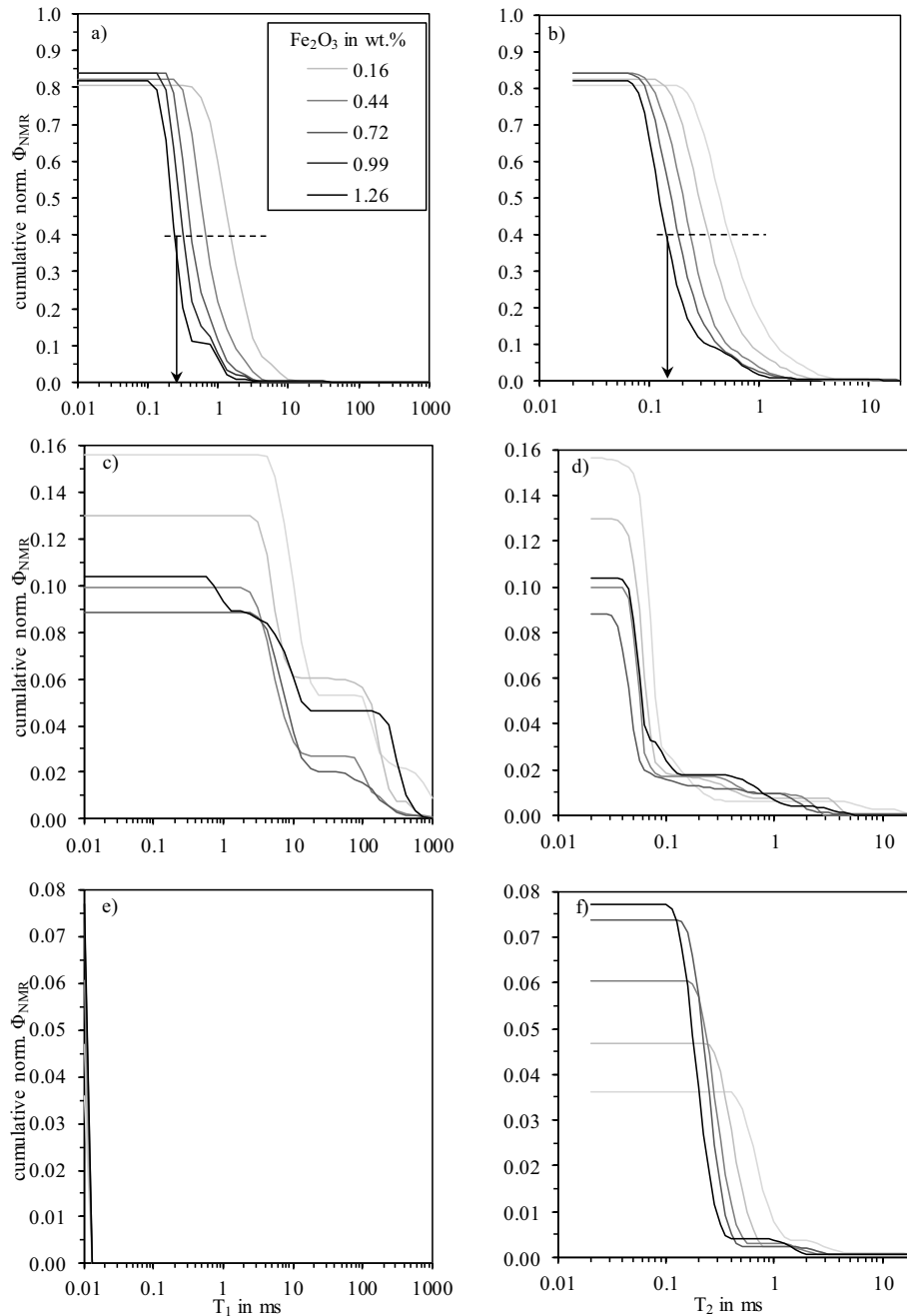


Fig. 9. T_1 and T_2 projections a) and b) of the signal along the $T_1 = T_2$ diagonal, presumably liquid water inside the porous hydrated calcium silicate structure, c) and d) of the off-diagonal signals, and e) and f) of the signal at shortest T_1 accounted for in the ILT evaluation at $T_1 = 0.01$ ms.

though slightly tilted in its T_1/T_2 alignment, and one OPC neatly following the diagonal though shifted to $T_1/T_2 = 2$. Based on the linear approximation in Fig. 10, we found an extrapolated T_1/T_2 of ~ 5 for a Fe_2O_3 free cement. Considering the influence of signal at shortest T_1 (see Fig. 9e) and f)) and adding this to the on-diagonal signal in this evaluation, this ratio is even slightly increased. For hydrated synthetic Alite, we found an averaged T_1/T_2 of roughly 7, matching this. For increasing amounts of Fe_2O_3 , T_1/T_2 converges to ~ 1.5 . The latter matches the results from McDonald and Co-workers [4,24],³ as well as a study on unhydratable calcium silicate (pseudo-Wollastonite), doped

with increased amounts of iron [6]. This interaction of paramagnetic and non-magnetic relaxation mechanisms is further discussed in Section 4.3. Assuming the shortest relaxation component found in the synthetic Alite at $T_2 = 0.5$ ms is associated with CSH interlayer water with a (simplified planar pore) size of ~ 0.9 nm [69], the surface relaxivity of hydrated calcium silicate surfaces $\lambda_{T_2, \text{CSH}}$ equates to ~ 0.9 nm/ms.

Concerning the larger water-filled pores, no T_2 relaxation appears at $T_2 > 20$ ms. T_2 of > 10 ms are rarely seen in healthy cementitious materials [29], even for higher water cement ratios [70] and might be reserved for drying induced [53,71] or deterioration porosity changes [72] as well as solvent saturated cement stone [73]. One explanation for its absence would be the self-desiccation and shrinkage of cement stone while hydration [74,75]. In that case, no larger pores would be found [21]. A reason might also be found in the saturation of pore

³ Considering the influence of higher NMR measurement frequency, slightly increasing T_1/T_2 ratio.

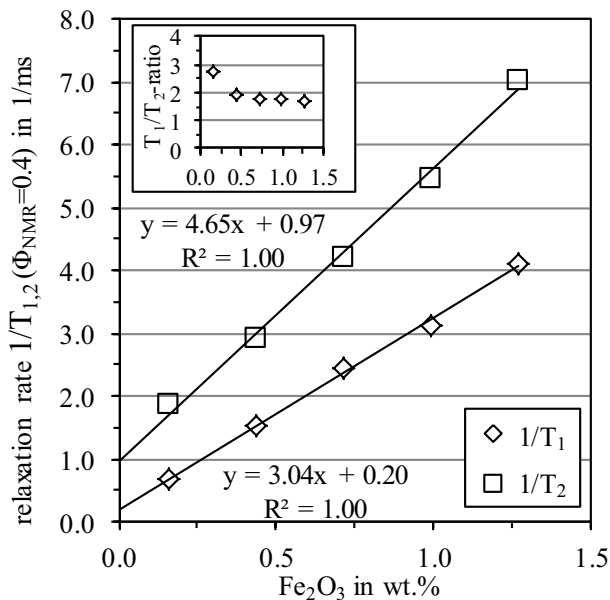


Fig. 10. Relaxation rate ($1/T_{1,2}$) for liquid water lying along the diagonal T_1 – T_2 at a 0.4 of the normalised NMR signal. The data is linearly approximated. The inset shows the T_1/T_2 ratio.

solution with iron hydroxide [5] increasing bulk relaxation with the high pH-values considerably increasing iron solubility [76].

The evaluation of results by ILT was done using a stronger smoothing and thus, provoking overlapping of the otherwise well discernible peaks for interlayer, gel, interhydrate, and capillary water described in the literature [20,69]. Other than appearing in Fig. 7 and Fig. 9, in a less severe regularized evaluation, these single peaks become separate and well distinguishable.

4.2. Influence of Fe_2O_3 on NMR relaxation of differently bound water

Signal from solid contents is only partly visible at very short T_2 , though spreading over a wide range of T_1 (see Figs. 4 and 7) as off-diagonal peaks. Differentiation of these and further signals from liquid water only becomes possible by applying different pulse sequences than CPMG, such as solid echo [17], or a combined T_1 – T_2 study as used here. As discussed further in Section 4.4, results from hydrated phases obtained on single-sided NMR by 2D-ILT are difficult to evaluate in a quantitative manner. Nonetheless, only the latter allows evaluation of data without a priori assumptions on the existing relaxation components, and thus, allowed the detection of further relaxation components at higher T_1/T_2 .

Considering the signal intensities of the powdered samples in Table 4 as well as the amounts of solid phases as found by XRD and TGA in the hydrated cement stones in Table 3, signal of ettringite, calcium hydroxide, and monocarboaluminate or hydrotalcite might be expected to be found in cement stone. The signal strength of these can be estimated using results in Section 3.2 as well as the density of these constituents [77] to roughly $\Phi_{NMR} = 0.5\%/wt\%$ and $0.2\%/wt\%$ for ettringite and calcium hydroxide, respectively.

As the major off-diagonal feature, we found a peak at $T_1/T_2 = 10\text{ ms}/0.075\text{ ms}$, most prominent for the WPC stone in Fig. 7b), as has been reported before [4,23,26]. This feature persists even after drying, as seen in Fig. 7a) and thus, cannot result from an exchange between pores of different sizes [4], exclusively. Based on our results from powdered substances in Fig. 4, we assign this peak to phases with aluminate rich surfaces, and predominantly ettringite. Thus, both NMR and XRD results agree on a reduction of ettringite for the OPC stone, though not on its extent in absolute numbers. Furthermore, in the fairly

regularized results presented in Fig. 7, the peak is divided into two for OPC with a second peak moving to smaller T_1 . This movement might be caused by the formation of further amorphous (and thus not visible to XRD) aluminate hydrogarnets, as might occur when only reduced amounts of aluminate are bound in ettringite and monocarboaluminate. Another explanation is a partial contribution of pore exchange to this peak as originally proposed [4], being erased by drying and on the other hand, moving along with the signal along the diagonal to shorter T_2 for higher Fe_2O_3 . As further discussed in Section 4.4, a final clarification on this matter requires the use of other ILT algorithms. Further peaks emerge for the samples at greater T_1 , indicating the existence of further constituents such e.g. portlandite and hydrotalcite though both appear hardly quantifiable based on the results presented here. Last, smaller off-diagonal peaks emerge for longer T_2 relaxation times with a possible explanation described in Section 4.3.

The original observation of a T_1 – T_2 off-diagonal peak [4,24,26] was closely connected to the finding of off-diagonal T_2 – T_2 exchange peaks and the findings here might also raise the question for re-interpretation of such past T_2 – T_2 results. It might be noted that some authors report a close connection between T_1 – T_2 and T_2 – T_2 measurements [4,42,48,78] with T_1 being a great influence on T_2 – T_2 distributions. The existence of an off-diagonal peak due to ettringite with considerable greater T_1/T_2 ratio as the signal along the diagonal might give rise to artefacts in the T_2 – T_2 distribution. If on the other hand no T_1 – T_2 off-diagonal peak exists [24], T_2 – T_2 off-diagonal peaks [26] can only be interpreted as exchange.

The existing relaxation time to pore-size conversion factor ($\lambda = 3.73\text{ nm/ms}$) based on the fast-diffusion model [20,21] was developed for WPC samples, based on a shortest T_2 relaxation of 0.075 ms found upon drying. We found exactly this T_2 relaxation time for powdered ettringite and the existing 2D T_1 – T_2 measurements for the closed magnet setup [4] indicate that this off-diagonal peak, formally assigned to pore water exchange, is well picked up by a CPMG sequence on this device. Furthermore, severe deformations in the sheet-like CSH structure were found while drying and rewetting [53,71] raising the question if upon drying more of a double-layered calcium silicate sandwich surrounds a single water layer rather than a single layer of water on the surface as needed for calibration of the fast diffusion model. Nonetheless, the existing relaxation time to pore size factor has proven to give consistent results, compared to numbers in the literature [20,21]. For dried WPC stone, we found a peak at 0.25 ms/0.075 ms that in 1D T_2 measurements would overlap with the more prominent ettringite peak. Nevertheless, it might be worth reinvestigating this matter, especially when aiming to find consistent pore sizes with other measurement techniques [21].

The results of hydrated WAC offer an abundance of discernible and reproducible features, the interpretation of which would lead beyond the scope of this work. It might be noted that due to temperature dependent hydration kinetics [79] as well as subsequent conversion effects [31] different CAH compositions form, which might become well discernible in the future using 1H NMR.

4.3. A theory of relaxation contributions in cement stone

Theories on NMR relaxation are numerous and in the following, we aim to provide a possible explanation for our results. Following the ideas that (i) the total $T_{1,2}$ relaxation rate is merely a combination of its single relaxing components [3,7] (see Eq. (1)), that (ii) in high surface materials such as porous cement stone, surface effects dominate relaxation, and that (iii) every mineral surface has its own unique T_1 surface relaxivity [5], we interpret our results as water surrounded by chemically different pore surfaces. For this, the idea of a mineral's unique T_1 surface relaxivity needs to be expanded to having its own T_1 as well as T_2 relaxivity and thus, its own unique T_1/T_2 ratio.

The results presented here indicate that iron-free hydrated calcium silicate has a T_1/T_2 of 7 (see Fig. 5). Contamination with even small amounts of paramagnetic iron reduce the T_1/T_2 to ~ 1.5 (see Fig. 7f)).

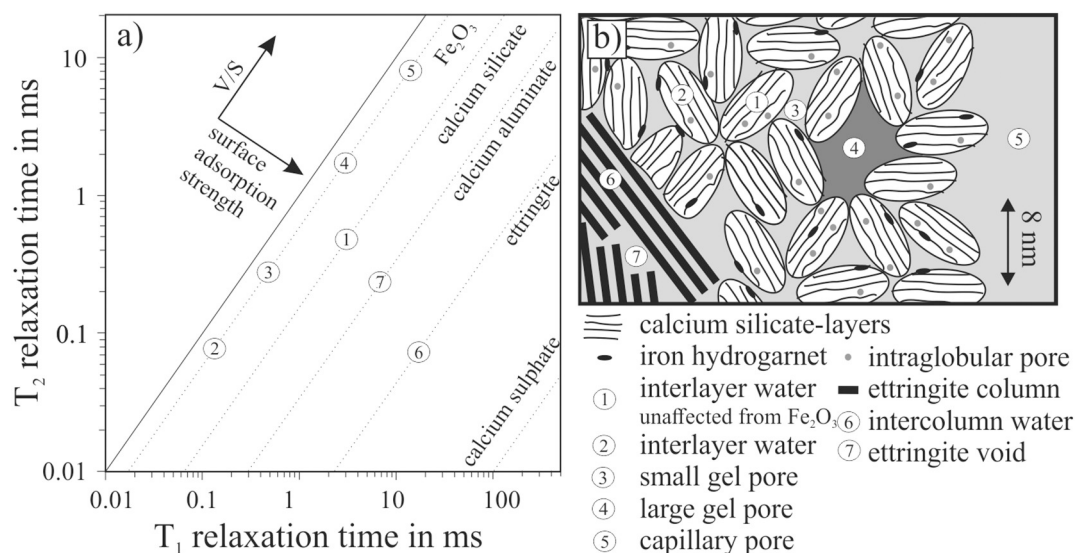


Fig. 11. Schematic representation of the theory of interacting relaxation mechanisms with a) the T_1 - T_2 relaxation times found in cement stone samples with the black line indicating $T_1/T_2 = 1$ (as found for SiO_2 [25] or unhydratable pseudo-Wollastonite [6]), the dotted lines indicate different T_1/T_2 ratios of ~ 1.5 for Fe_2O_3 rich surfaces, ~ 7 for Fe_2O_3 free hydrated calcium silicate, ~ 25 for calcium aluminate, ~ 133 for ettringite, and $\sim 10,000$ for calcium sulphate. b) Schematic representation of cement structure loosely based on Jennings' globular based CM-II model [22] (including intraglobular pores, interlayer water, small and large gel pores, and capillary water) with regard to a previous publication of the authors [53] with added iron hydrogarnet and ettringite column structure.

Aluminate rich surfaces show a greater T_1/T_2 , visible for hydrated WAC (in Fig. 6) though best seen for water-mixed ettringite powder (in the supplement) with a T_1/T_2 of ~ 25 . Last, sulphates appear to increase T_1/T_2 even further. In general, the fastest relaxing surface inside a well-connected pore regime dominates the overall relaxation (compare Eq. (1)) and masks all other contributions.

Concerning the interaction of calcium silicate and iron-bearing phases in cement stone, the highly structured interlayer water in hydrated calcium silicates might be partly free of and uninfluenced by iron. Relaxation then is controlled by calcium silicate surface interaction and thus, an increased T_1/T_2 is found. This is seen in the results for the cement samples presented here and elsewhere [4,24]. With increasing pore size, water becomes more mobile and the probability of encountering iron-containing phases increases. For larger pores and greater amounts of Fe_2O_3 , the T_1/T_2 closer resembles the ratio of these phases ($T_1/T_2 \sim 1.5$). A curvature in T_1 - T_2 relaxation distribution suggests that the distribution of iron-bearing phases is not necessarily homogeneous in hydrated cement and might depend on hydration kinetics and products formed. Furthermore, the reported change of T_1/T_2 from 4 to 2 for WPC and OPC [24] then indicates a change in relaxation contribution from predominantly calcium silicate surface relaxation to Fe_2O_3 dominated paramagnetic relaxation.

Moving up the $T_1 = T_2$ parallel is associated with an increase in pore size [3] (compare also Fig. 2). Signal with the same T_1/T_2 as e.g. calcium silicate, calcium aluminate, ettringite or gypsum though with higher relaxation times is emitted by liquid water, bound in a chemically similar environment, though in larger voids comparable to the size of e.g. gel pores. For CSH, this is well established. Ettringite forms micrometre-sized crystals, as suggested by TEM [31], making the assumption of the existence of nanometre-sized voids with a calcium aluminate enriched surface inside it quite reasonable. Consequently, the small off-diagonal peaks at longer T_2 relaxation might arise from such larger voids in ettringite (or similar structures) forming micrometre sized domains inside hydrated cement paste. Fig. 11 depicts the assumed structure of differently sized pores in CSH as well as column-shaped ettringite [30], together with the corresponding T_1/T_2 pairs. Depending on the connectivity of these chemically different structures, these might become discernible in cement paste. Even small amounts of Fe_2O_3 would mask such effects due to its great influence on relaxation.

Partial drying of samples might reduce connectivity here and thus, allow better separation, as seen in Fig. 7a).

4.4. Limitation and possibilities of the NMR techniques used

We performed complex 2D T_1 - T_2 measurements on a single-sided NMR device, naturally working in a highly inhomogeneous magnetic field. The evaluation was done by a sophisticated though still continuously improved 2D-ILT algorithm. We found that overall peak sums as well as peaks ratios of 2D-ILT distributions vary with regularization parameter α , which becomes apparent for simple bulk water experiments (compare Fig. 4 and Table 4); an unfortunate circumstance, which limits the interpretation of results in absolute terms. Furthermore, the relaxation of solid bound hydrogen is usually described by a Gaussian behaviour [18,19] and the chosen exponential based evaluation induces further alteration to T_2 and its amplitudes, as theoretically tested (not shown). All this undeniably leads to the drawback that unlike signal from liquid water [53], the signal from solid cement phases cannot be interpreted quantitatively in single-sided ^1H NMR 2D-ILT experiments. The implementation of improved regularization [63], a combined exponential and Gaussian ILT evaluation [80] as well as substance dependent calibrations might improve interpretation. Another major issue that needs to be addressed in future studies is the existence of negative exchange peaks [42,44,45] in 2D T_1 - T_2 measurements. As shown by its persistence after drying, the off-diagonal peak cannot be assigned to pore-exchange [4,24,26] alone, but a solid signal component such as ettringite. None the less, we cannot be certain as to what extent further peaks are the consequence of exchange with counter negative exchange peaks being neglected. More elaborate inversion algorithms [46,47] might elucidate this further.

The very short echo spacing of 30 μs allows the registration of, at least, traces of solid signal. However, the short echo spacing paired with a pulse length of 4.5 μs , leads to spin-locking effects that are expected to be rather prominent in our measurements and might be more severe than the reported deviations of 20% [49,50]. These increase the apparent T_2 , and thus, bringing it closer to its corresponding T_1 . These effects are most pronounced for a signal with large T_1/T_2 . This might explain the detection of overall longer T_2 for solids compared to the numbers in the literature [17]. Lastly, the signal-to-noise ratio in single-

sided experiments [81] is not as good as for closed magnet setups. All in all, systematic studies in closed magnet setups, more controllably exciting spins with considerably better signal-to-noise ratios, variable echo-spacing reducing spin-lock, short spectrometer dead times, and most importantly, attainable free induction decays in homogeneous magnetic fields might provide further insight.

4.5. Limitation of this study due to the cement samples used

The cements used in this study were chosen for their good comparability in terms of its elemental constituents (expressed in oxides) with only the amount of Fe_2O_3 differing. Mixing these two cements will lead to a more heterogeneous distribution of constituents in the resulting cement powder, compared to a combined burning [82]. It should be kept in mind that in every cement, sulphate setting regulator or in other cases granulated slag or limestone are only added after burning and while grinding, causing heterogeneity on the particle length scale. Furthermore, even burned clinker may retain clinker crystal domains in the micrometre range (visible e.g. in [83]). At least part of this heterogeneity is reduced in the solution-precipitation process while hydration. Our results of linear increase of $T_{1,2}$ relaxation rate due to Fe_2O_3 content are best explained based on the known relationship of the two parameters assuming an overall good homogeneity of Fe_2O_3 . Furthermore, other than in past studies where inert powders were mixed to incorporate increasing amounts of iron [8], a reactive interaction between iron-bearing clinker phases and other cement constituents occurs. More precisely, based on Dilnesa et al. [33], a conversion of Brownmillerite to ferrihydrite and subsequently iron bearing silica hydrogarnet $\text{C}(\text{A},\text{F})_2\text{SH}$ is expected. Results are interpreted in terms of comparative homogeneously distributed iron-bearing hydrogarnet, though evenly precipitated ferrihydrite might as well be the prominent iron phase. A locally concentrated occurrence of iron hydrogarnet might increase relaxation of bound water to an extent that signal is not detectable by single-sided ^1H NMR anymore. For our samples, XRD results show that all Brownmillerite reacted, thus allowing the use of the cement's original Fe_2O_3 content as an indicator of the amount of paramagnetic impurities influencing relaxation. The OPC used in our study contains only a limited amount of iron, with other OPC incorporating Fe_2O_3 amounts of up to 8 wt%. It should be highly questioned that the conversion of relaxation rate in dependence of Fe_2O_3 in the unreacted clinker (compare Fig. 10) holds for such cements. Only a limited and slow hydration reaction of Brownmillerite is expected which strongly interacts with the hydration of calcium aluminate clinker. Unhydrated Brownmillerite, by definition, is not in contact with water and thus, does not contribute to relaxation accordingly. Lastly, ascribing changes in relaxation between white and grey cements solely to the amount of Fe_2O_3 neglects the influence of other paramagnetic species, such as MnO.

5. Conclusions

There is an increasing interest in the determination of porosity and pore size distributions by ^1H NMR relaxometry. Based on our results, we conclude:

- Paramagnetic impurities, such as Fe_2O_3 , linearly increase both T_1 and T_2 surface relaxation of liquid water in cementitious materials. This might be used for an adjusted factor for the conversion of relaxation time to pore size for OPC.
- Besides detecting liquid water, 2D T_1 - T_2 measurements allow the separation of further signal. The latter appears to be not directly influenced by paramagnetic impurities itself, though gives information on the hydration products formed. The most prominent off-diagonal peak has been attributed to ettringite. Further peaks are found, best explained by chemically different bound liquid water. We present a theory of contributing relaxation mechanisms,

explaining the complex 2D relaxation behaviour found for cement stone samples.

- Both the used 2D-ILT algorithm as well as the single-sided NMR device currently limit the quantitative evaluation of results, especially for fast relaxing signals. Using closed magnet setups and improving evaluation, ^1H NMR might provide quantitative information on all hydration products and thus, a complete picture of the hydrated nanostructure of cementitious materials.
- Single-sided ^1H NMR might prove itself to be useful for qualitative non-destructive depth-resolved evaluations of hydrated phases and porosity changes in cover concrete for future research on e.g. leaching, acid or sulphate attack.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.cemconres.2019.03.027>.

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